

Hyungcheol Shin

其他

基本信息

姓名: Hyungcheol Shin

所属院校: 首尔国立大学

地区: South Korea

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

微电子器件与制造

纳米电子与先进工艺

存储器设计与架构

个人简介

Dr. Hyungcheol Shin is a distinguished professor at Seoul National University's Department of Electrical and Computer Engineering. His research focuses on CMOS logic technology, DRAM cells, and NAND flash memory. Dr. Shin has made significant contributions to the field through his publications on three-dimensional NAND flash architecture, single-crystalline Si stacked array (STAR) NAND flash memory, and highly functional STT-MRAM. His work has been widely cited, showcasing his impactful research in semiconductor technology and device engineering.

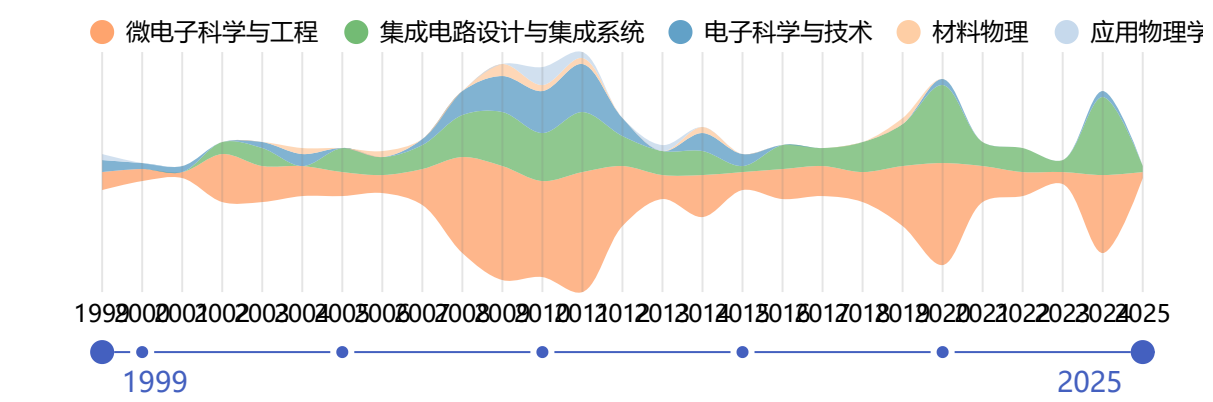
工作经历

Seoul National University

教育经历

Seoul National University

近年发表的论文



Modeling of effective mobility in 3D NAND flash memory with polycrystalline silicon channel

JOURNAL OF COMPUTATIONAL ELECTRONICS

Kim, Juhyun; Shin, Hyungcheol

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Modeling of Threshold Voltage Shift by Neighboring Transistors for Macaroni Channel MOSFETs in Series

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Nguyen-Gia, Quan; Shin, Hyungcheol

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Machine Learning-Based Prediction of Threshold Voltage Distribution Due to Lateral Migration in 3-D NAND Flash Memory

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Han, Insang; Lee, Jang Kyu; Ahn, Sangmin; Shin, Hyungcheol

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A New Semi-Analytical Model for Erase Transients of 3-D Gate-All-Around (GAA) NAND Flash Memories

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Yoo, Jinil; Choi, Haechan; Shin, Hyungcheol

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Time Constant Analysis of Lateral Charge Loss in 3-D NAND Flash Memories Through Multiscale Simulations

IEEE TRANSACTIONS ON ELECTRON DEVICES

Yoo, Jinil; Jo, Hyungjun; Shin, Hyungcheol

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A New Physical Model for Program Transients of Cylindrical Charge-Trap-Based NAND Flash Memories

IEEE TRANSACTIONS ON ELECTRON DEVICES

Choi, Haechan; Yoo, Jinil; Shin, Hyungcheol

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Investigation of Endurance Characteristics in 3-D nand Flash Memory With Trap Profile Analysis

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Jo, Hyungjun; Kim, Jongwoo; Cho, Yonggyu; Shim, Hyunyoung; Sim, Jaesung; Shin, Hyungcheol

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IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

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JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Cho, Seongjae; Lee, Sung-Tae; Kim, Soomin; Shin, Hyungcheol

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First demonstration of SRAM transistor based on 3-dimensional stacked FET with back side interconnection structure beyond 1nm node

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Kim, Mingyu; Park, Jaehyun; Park, Sungil; Park, Jejune; Kim, Juhyun; Ha, Daewon; Shin, Hyungcheol

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A Compact Model for Program Operation of Gate-All-Around Barrier-Engineered Charge-Trapping NAND Flash Memories

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IEEE TRANSACTIONS ON ELECTRON DEVICES

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Distinguishing capture cross-section parameter between GIDL erase compact model and TCAD

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Analysis of the Effect of Residual Holes on Lateral Migration During the Retention Operation in 3-D NAND Flash Memory

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Characteristic Length of Macaroni Channel MOSFET (vol 40, pg 1720, 2019)

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IEEE TRANSACTIONS ON ELECTRON DEVICES

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Analysis of Failure Mechanisms During the Long-Term Retention Operation in 3-D NAND Flash Memories

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Variability-Aware Machine Learning Strategy for 3-D NAND Flash Memories

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Analysis on Temperature Dependence of Hot Carrier Degradation by Mechanism Separation

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New Insight Into Negative Bias Temperature Instability Degradation During Self-Heating in Nanoscale Bulk FinFETs

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Prediction of Process Variation Effect for Ultrascaled GAA Vertical FET Devices Using a Machine Learning Approach

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Strain Engineering for 3.5-nm Node in Stacked-Nanoplate FET

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Modeling of Nanoplate Parasitic Extension Resistance and Its Associated Dependency on Spacer Materials

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Investigation of Gate Sidewall Spacer Optimization From OFF-State Leakage Current Perspective in 3-nm Node Device

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Fringe Capacitance Modeling in NanoPlate MOSFET Using Conformal Mapping

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Jongsu; Kang, Myounggon; Jeon, Jongwook; Shin, Hyungcheol

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Compact Model Strategy of Metal-Gate Work-Function Variation for Ultrascaled FinFET and Vertical GAA FETs

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Thermal-Aware Shallow Trench Isolation Design Optimization for Minimizing IOFF in Various Sub-10-nm 3-D Transistors

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Analysis and Modeling of Program Disturbance by Hot Carrier Injection in 3D NAND Flash Memory Using TCAD

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

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SOLID-STATE ELECTRONICS

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2018

Comparison of dual-k spacer and single-k spacer for single NWFET and 3-stack NWFET

SOLID-STATE ELECTRONICS

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2018

Analysis of electrical characteristics and proposal of design guide for ultra-scaled nanoplate vertical FET and 6T-SRAM

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Comparison of work function variation between FinFET and 3D stacked nanowire FET devices for 6-T SRAM reliability

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Analysis of Electrothermal Characteristics of GAA Vertical Nanoplate-Shaped FETs

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Impact of Trap Position on Random Telegraph Noise in a 70-Å Nanowire eld-Effect Transistor

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Lee, Hyunseul; Cho, Karam; Shin, Changhwan; Shin, Hyungcheol

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Improvement of Characteristics with a Sub-5 nm Ge-Doped Silicon Nitride Layer in Charge Trap Flash Memory Cells

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Li, Dong Hua; Kim, Wandong; Shim, Won Bo; Park, Se Hwan; Kim, Yoon; Lee, Gil Sung; Kim, Doo-Hyun; Lee, Jung Hoon; Yun, Jang-Gn; Cho, Seongjae; Park, Il Han; Lee, Jong-Ho; Shin, Hyungcheol; Park, Byung-Gook

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Characterizing traps causing random telegraph noise during trap-assisted tunneling gate-induced drain leakage

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JAPANESE JOURNAL OF APPLIED PHYSICS

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Step-by-step implementation of an amplifier circuit with a graphene field-effect transistor on a printed-circuit board

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Is quantum capacitance in graphene a potential hurdle for device scaling?

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Probability Level Dependence of Failure Mechanisms in Sub-20 nm NAND Flash Memory

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Separation of Corner Component in TAT Mechanism in Retention Characteristics of Sub 20-nm NAND Flash Memory

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Analysis and modeling for random telegraph noise of GIDL current in saddle MOSFET for DRAM application

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Patterning of Si nanowire array with electron beam lithography for sub-22 nm Si nanoelectronics technology

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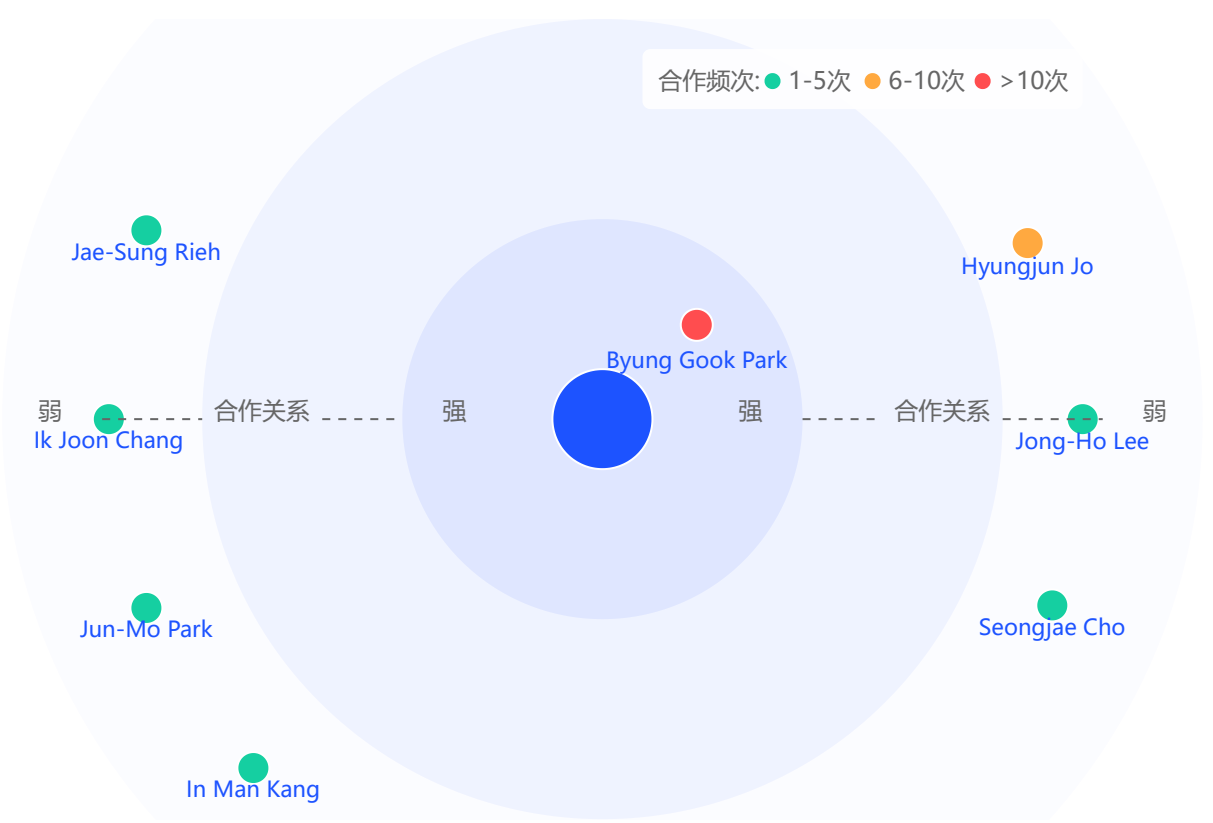
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